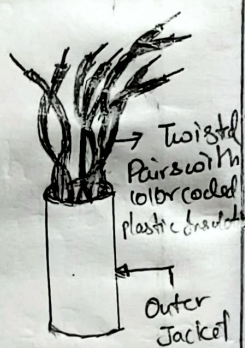
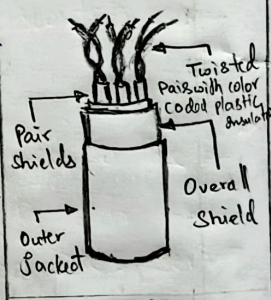
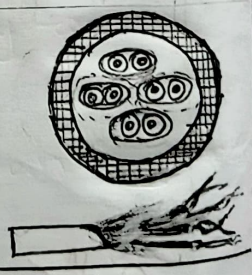
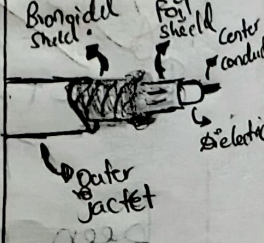
Ex: No: 02
23/07/26

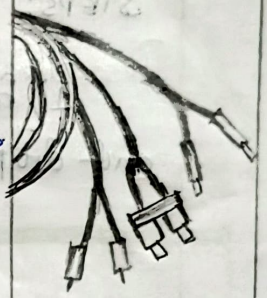
STUDY OF NETWORK CABLES

AIM: Study of different types of network cables

DIFFERENT TYPES OF NETWORK CABLES:

1) Unsheilded Twisted Pair (UTP) cable
2) Sheilded Twisted Pair (STP) cable
3) Coaxial Cable.
4) Fibre Optic Cable.

Cable Type	Category	Maximum Data Transmission	Advantage/ Disadvantage	Application/ Use.	Image
UTP	Category 3	10 bps	Advantage: * cheaper in cost. * Easy to install as they have a smaller overall diameter. Disadvantage: * More prone to EMI and noise	10 Base-T Ethernet	
	Category 5	Upto 100 Mbps		Fast Ethernet, Gigabit Ethernet.	
	Category 5e	1Gbps		Fast Ethernet, Gigabit Ethernet	
STP	Category 6, 6a.	10Gbps	Advantage: * Shielded * Faster than UTP. * Less susceptible to noise and interference Disadvantage * Expensive * Greater installation Effort	Gigabit Ethernet 10G Ethernet (55m) widely used in data Centres.	
SSTP	Category 7	10Gbps		Gigabit Ethernet, 10G Ethernet (100m)	
Coaxial Cable	RG-6 RG-59 RG-11	10-100 Mbps	Advantage: * high bandwidth * Immune to interference * Low loss bandwidth	Speed of signal is 500m Television network High Speed Internet connection	

			* Long distance Disadvantages * Expensive * Required skilled Installers		
Fibre optics Cable.	Sigle mode Multi mode	100Gbps	Advantages * High speed * High bandwidth * High Security * Long distance Disadvantages * Expensive * Required Skilled Installers	Maximum distance of fiber optics cable is around 100 meters	

RESULT:

The types of Network Cables and making Cross / coierd Cable & straight through Cable using Clamping (crimping tool.